Quantifying and Attributing Polarization to Annotator Groups

Dimitris Tsirmpas[1,2] and John Pavlopoulos[1,2*]

[1]Department of Informatics, Athens University of Economics and Business, 76, Patission, Athens, GR-10434, Attiki, Greece.
[2]Archimedes Unit, Athena Research Center, 1, Artemidos, Athens, GR-15125, Attiki, Greece.

*Corresponding author(s). E-mail(s): annis@aueb.gr;
Contributing authors: dim.tsirmpas@aueb.gr;

Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: Polarization, annotation, annotators, socio-demographic, metric

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields such as content moderation, toxicity detection and hate speech detection. These tasks are especially essential for training systems that protect vulnerable and minority groups in online spaces, where they are often targeted [1, 2]. However, annotations for such tasks are subjective and usually based on majority-group annotators. This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems.

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree in vain. An example of such cases would be racist comments that are presented with coded language (usually referred to as "dog-whistles" [3]), which are

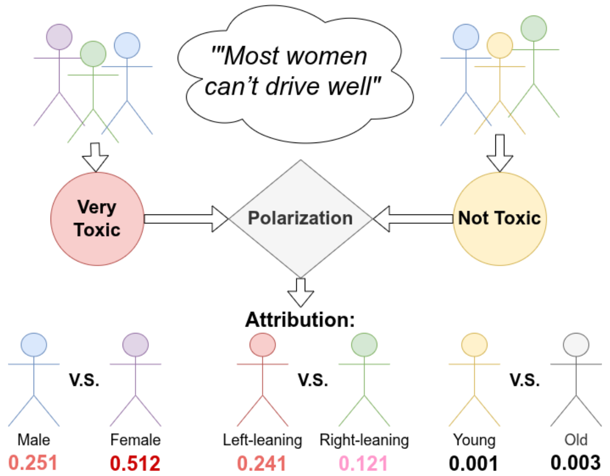

Fig. 1: The Apunim framework attributes annotator polarization to certain characteristics.

not picked up by most annotators, but only by ones belonging in the targeted minority. Inversely, majority-group annotators may bias their own annotations against the minority-group annotators. [4] for instance show that toxicity models disproportionally marked African American speech as "toxic".

Inter-annotator agreement is often used to detect such cases [5]. Metrics based on agreement mostly measure annotation variance, which may not be an optimal problem formulation for detecting such cases. *Polarization* is a better instrument in this regard, since it formulates annotation analysis as a cluster identification problem [6, 7]—specifically whether two or more annotation clusters are present (Figure 2).

While we can detect polarization [6, 7], there is little work on how to detect whether it is actually caused by marginalized groups. [7] propose a mechanism which only works in the extreme case where overall polarization is zero; in this work we demonstrate that this case is not frequent in real-world data. We instead propose the *Apunim ("Aposteriori Unimodality") metric*, which attributes polarization to individual annotator sociodemographic groups. Our contributions are:

1. We introduce a new metric which measures how much polarization can be attributed specific sociodemographic groups.
2. We discover issues not yet acknowledged in literature, such as the presence of inherent polarization and conflicting polarization directions. Our proposed metric is designed to avoid these issues.
3. We apply our metric to four datasets; two with human comments and annotations [8, 9] as well as two generated and annotated by Large Language Model (LLM) agents [10].

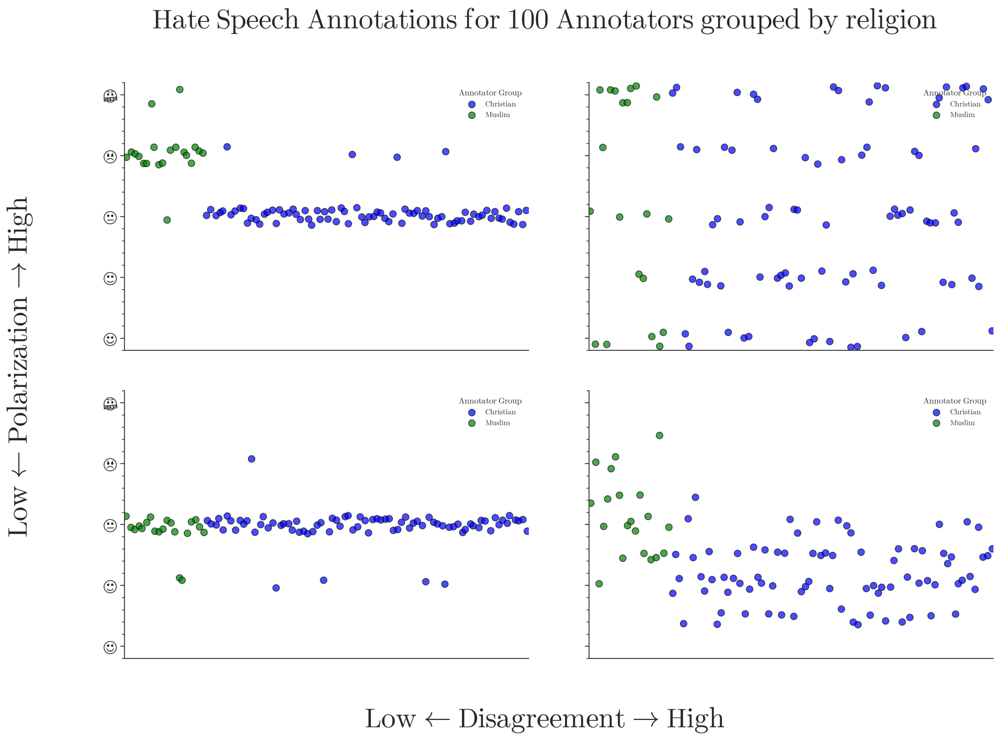

Fig. 2: Disagreement is similar to variance; thus it overlooks cases where annotators may be "split down the middle". Polarization on the other hand, is able to identify multiple clusters even if they are close together.

2 How to attribute polarization

2.1 Problem Formulation

Sociodemographic Backgrounds

Since our goal is to pinpoint which specific characteristics contribute to polarization, we need to isolate individual groups within a SocioDemographic Background (SDB). Let Θ be one of multiple "dimensions" the set of all annotator SDB groups for a single dimension (e.g. "male", "female" and "non-binary" if we are investigating annotator gender).

Annotations

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated datapoints.[1] These datapoints could be of any modality, such as comments in a discussion, images, videos or survey questions. We assume that how polarizing a datapoint is depends on two

[1] We use the general term "datapoint" instead of domain-specific terms, such as "comments", even though the examples presented in the paper are comments in discussions.

variables: *inherent ("apriori") polarization*, and polarization caused by *different annotator SDBs ("aposteriori" polarization)*. Assuming that each datapoint c is assigned multiple annotators, we can define the annotation multi-set $A(c) = \{(a, \theta)\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task grouped by gender, would be formulated as:

$$A(\textit{``Could be better, could be worse''}) = \{(+, F), (—, M), (—, F), \ldots\} \tag{1}$$

Polarization

Each set of annotations exhibits a certain degree of *polarization*. In the absence of polarization, annotation distributions are typically unimodal: annotators may disagree on finer details—shown as a bell-shaped distribution around a central mode—rather than on fundamentally different interpretations, which would instead give rise to multimodal distributions. To our knowledge, the only polarization metric explicitly grounded in this assumption is normalized Distance From Unimodality (nDFU) [7]. The metric takes values in the [0, 1] range, with $nDFU \approx 0$ indicating no polarization (unimodal distributions) and $nDFU \approx 1$ corresponding to complete polarization (highly multimodal distributions). Our objective is therefore to quantify the extent to which the $nDFU$ of a polarizing datapoint can be partially explained by a given SDB group θ.

2.2 Intuition

Polarization attribution

Intuitively, θ partially explains the polarization of a datapoint c when the annotations grouped by θ are more polarized compared to the full set of annotations. Figure 3 exhibits a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators. The annotations are generally polarized ($nDFU_{all} = 0.625$). The annotations by female annotators exhibit low polarization between themselves ($nDFU_{women} = 0.1$), since most agree the data-point is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.3725$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by disagreements between male and female annotators.

Previous work [7] stopped on a datapoint-level analysis. In practice however, this kind of analysis is inherently noisy due to a lack of information (i.e., annotations) relating to each individual datapoint. In order to get meaningful information, we need to consider information *between* datapoints, as the number of datapoints is usually several orders of magnitude larger than the number of annotations per datapoint (§3). In other words, instead of investigating whether a specific group becomes polarized by an individual datapoint (e.g. comment), we investigate whether that group *generally becomes polarized on the specific dataset and task.*

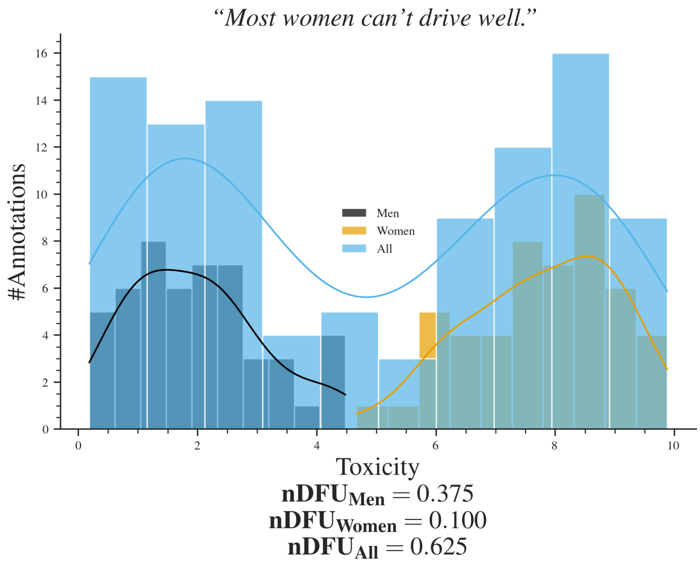

Fig. 3: Example of a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

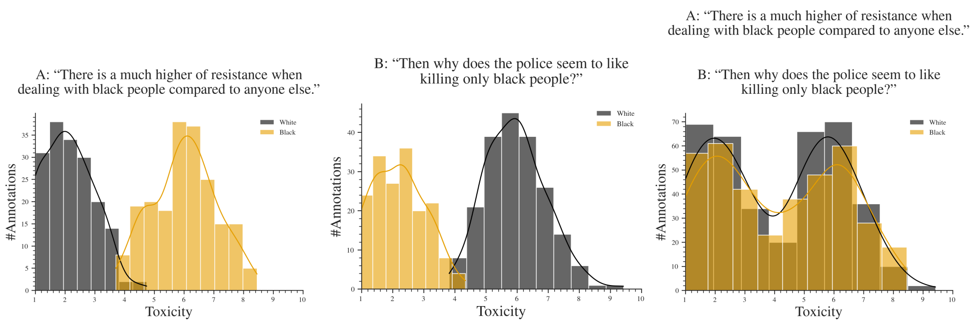

Fig. 4: Hypothetical example of a polarizing discussion with two comments, for which the annotators disagree on which is the toxic one. If we aggregate the two comments, the polarization scores for both groups significantly rises (we need to compare two bimodal distributions), obscuring whether the exhibited polarization can be partially attributed to ethnic background.

Dataset polarization

Given this observation, we would be tempted to quantify dataset-level polarization attribution by simply aggregating all annotations in the dataset and comparing the nDFU of each θ with the full set of annotations $A(c)$. In fact this is very tempting since aggregating annotations solves the very real and important problem of annotation scarcity; each datapoint may be annotated by only a handful of annotators (5 is a common number in many publications), leading to a best-case even split between groups (for 5 annotators, a best case scenario would be only two groups with a 3-2 split). Since nDFU is at its core a histogram estimation, 3 annotations may not be enough to acquire a reliable histogram, and thus estimate of polarization. This is discussed in more depth in §5.

However, arbitrarily combining annotations would not work well in practice. Figure 4 illustrates a hypothetical discussion with two comments, both of which are toxic, but where the two groups of annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This example demonstrates that dataset-level polarization has a *direction*, which is not captured by the (symmetric) nDFU metric.

2.3 Aposteriori polarization

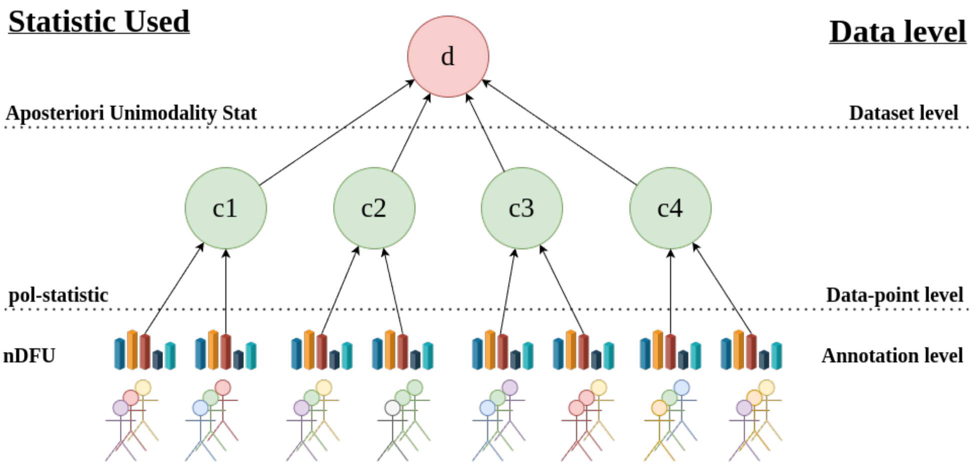

Fig. 5: An overview of how annotation information is sequentially aggregated for the *Apunim* metric. We gather statistical information from the annotations (different SDBs are denoted by different colors) via the nDFU measure, which is aggregated by the $pol_{apriori}$ and $pol_{observed}$ statistics for each datapoint. We aggregate these statistics on the dataset level, and compare them to get a chance-corrected *Apunim* value.

How polarizing a datapoint $c \in d$ is for annotators of group θ can be computed as $nDFU(P(A(c), \theta))$, where $P(A(c), \theta_i) = \{a(c; \theta) \in A(c) | \theta = \theta_i\}$ is the set of annotations for c from annotators belonging to the SDB group θ_i.

In §1 we mentioned that annotator polarization can be caused either by the annotators SDB, or other factors. The latter can be considered "apriori" polarization, which is driven by the inherent "controversy" of the data point. Previous work [7] considered the case where this apriori polarization was zero; this is not the case in many datasets, as we demonstrate in §3.

Unfortunately, we can not directly follow the intuition presented in §2.2 and compare the differences in nDFU between $A(c)$ and $P(A(c), \theta)$, as we would be comparing statistics between a subset and its superset (see Appendix B). Comparing the polarization of groups between themselves is not an option; if we compared the two annotator groups in Figure 3 we would conclude that there is no polarization present, since they have extremely similar nDFUs. In a scenario in which each datapoint was annotated with a very large amount of annotations, we could use bootstrapping to compare the apriori and aposteriori polarization for each comment. This however is an unrealistic assumption, since the costs of human annotation force researchers to use few annotators for each task [11].

To get around this limitation, we randomly partition the annotations in $|\Theta|$ groups, with matching group sizes,[2] which effectively "breaks up" the effect of any single isolated group on the polarization of the sample. The randomly partitioned annotations are then sampled t times. Formally, the average apriori polarization for a single datapoint will be given by:

$$\frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{2}$$

where $\tilde{P}_i$ is the random partition operator with matching group sizes.

While the low number of annotators is an important limitation when acquiring the observed polarization [11], this is decisively not the case with the number of datapoints themselves which can range up to hundreds of thousands. Thus, we have the luxury of filtering them before computing a dataset-wide metric. Since our metric attributes polarization, unpolarizing datapoints are useless. The same applies for datapoints which are composed of only a single annotator group. We define the subset S_d of dataset d such that:

$$S_d = \{c \in d \mid nDFU(c) > \alpha \textit{ and } c \textit{ annotated by at least two groups}\} \tag{3}$$

where α is a small positive number.

We can estimate how much polarization is generally present in the dataset, irrespective of any SDB by simply averaging the *apriori* polarization of each datapoint:

$$pol_{apriori}(d) = \frac{1}{|d|}\sum_{c \in S_d}\frac{1}{t}\sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c))) \tag{4}$$

Intuitively, a high apriori polarization value means that much of the polarization in the dataset is driven by factors other than SDBs. In this case, we would need to observe very strong polarization within a specific SDB group in order to conclude that the overall dataset polarization can be attributed to that group. This value will be compared to the average polarization present in the dataset by annotators of SDB group θ:

[2] E.g., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we will create random partitions of sizes 80 and 20.

$$pol_{obs.}(d,\, \theta) = \frac{1}{|d|} \sum_{c \in S_d} pol(c,\, \theta) \tag{5}$$

This formulation implicitly assumes that the overall polarization of a dataset can be captured by averaging polarization scores across datapoints. We consider this assumption reasonable: the mean is a simple and intuitive operator, and it exhibits appropriate sensitivity to outliers. Given that polarization values lie in the [0, 1] range, a single extreme datapoint is unlikely to substantially affect the estimate of overall polarization. In contrast, the presence of many such extreme values will meaningfully increase this estimate, which is the pattern we want to capture.

The next question would be how to compare the two values in Equations 4, 5. Given the inherent "noise" associated with most annotation tasks, a direct subtraction would not be useful. Instead we define the chance-corrected *"Apunim" (Aposteriori Unimodality)* metric as:

$$apunim(d,\, \theta) = \frac{pol_{obs.}(d,\, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)} \tag{6}$$

The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 5. Since $nDFU \to [0,\, 1] \Rightarrow apunim \to [-1,\, 1]$. The interpretation of this metric boils down to three scenarios:

- If $apunim(d,\, \theta) \approx 0$, the exhibited polarization among annotators of group θ does not meaningfully differ from all other groups. In other words, it is indistinguishable from noise.
- If $apunim(d,\, \theta) > 0$, then the polarization observed in the annotation task rises when this group is added to the annotation pool (equivalently, intragroup polarization is less than intergroup polarization). Example: If we observe $apunim(d,\, Afr.American) = 0.3$, $apunim(d,\, White) = 0.001$ in a hate-speech detection task, it is likely that African American annotators fundamentally disagree with white annotators on what constitutes racism in the dataset d.
- Conversely, if $apunim(d,\, \theta) < 0$, then the overall polarization would actually *rise* had we removed this group from the annotation task. Mathematically, this occurs when *apriori polarization is higher than the observed polarization* (intragroup polarization is higher than integroup polarization). In other words, annotators of that group may be very polarized between themselves. Example: If we observe $apunim(d,\, Latino) = -0.3$, $apunim(d,\, White) = 0.001$ in a hate-speech detection task, it is likely that Latino annotators fundamentally disagree with *each other* (compared to African American or white annotators) on which texts in the dataset d constitutes racism. This may indicate that *other dimensions (acting as confounding variables) are exerting heavy influence on the sample.*

2.4 p-value estimation

The condition $apunim(d,\, \theta) \approx 0$ is not well defined in the presence of inherent noise, which is why we need a statistical test that checks whether the deviation from 0 is significant. In other words we need to implement the following test:

Algorithm 1 P-value estimation for aposteriori polarization

Require: Dataset d, annotator groups Θ
Require: Filtered datapoints S_d
Require: Number of random partitions t
Ensure: p-value for testing whether $apunim(d, \theta) \neq 0$

Observed polarization attribution

$pol_{obs.}(d, \theta) = \frac{1}{|S_d|} \sum_{c \in S_d} nDFU(P(A(c), \theta))$

$pol_{apriori}(d) = \frac{1}{|S_d|} \sum_{c \in S_d} \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{P}_i(A(c)))$

$obs\_apunim = \frac{pol_{obs.}(d, \theta) - pol_{apriori}(d)}{1 - pol_{apriori}(d)}$

Null distribution

$Apunim_{\text{null}} = \left\{ apunim^{(i)} \;\middle|\; apunim^{(i)} = \frac{\bar{D}^{(i)} - \bar{D}^{(-i)}}{1 - \bar{D}^{(-i)}}, \;\; i = 1, \ldots, t \right\}$

where $\bar{D}^{(i)}$ denotes the dataset-level mean polarization obtained from the i-th randomized partition.

Statistical test

$p = Student\text{-}T(Apunim_{\text{null}} \neq obs\_apunim)$

return p

$$H_0 : \forall \theta \in \Theta,\; apunim(d, \theta) = 0,\; H_i : apunim(d, \theta_i) \neq 0 \tag{7}$$

For each group, we construct a null distribution of the apunim statistic by treating each randomized partition (each element in Equation 4, but without averaging the result) in turn as if it were the observed grouping—essentially, we compute a pseudo-apunim value by comparing its mean DFU to the mean DFU of the remaining random partitions. Repeating this process across all randomizations gives us an empirical null distribution of apunim values that reflects polarization expected by chance under the same group-size constraints. The observed apunim statistic is then compared against this null distribution using a two-sided one-sample t-test. The resulting p-value quantifies how unlikely the observed group polarization is, under the null hypothesis that annotator groups do not contribute to polarization beyond chance. The p-value estimation algorithm can be found in Algorithm 1.

Our test assumes that there exist (1) enough data-points to reliably run the means-test, and (2) enough annotations in each datapoint and SDB group to estimate the nDFUs in Equations 5,4. The former can be safely assumed for most datasets, because they feature enough data-points to reliably assume normal residuals ($n \gg 30$—see §3), which is why we use the Student-T test [12]. We discuss the latter assumption in Appendix 5.

Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values. We choose the Holms method [13]. The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses [14]). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

3 Datasets

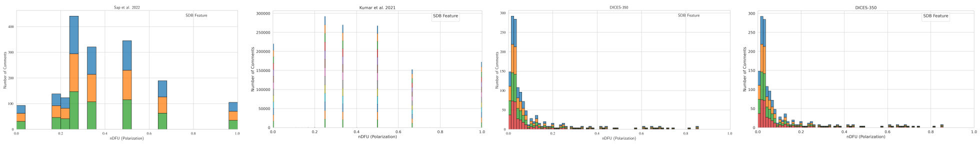

Fig. 6: Histogram of dataset polarization for the human datasets considered in this paper per SDB feature.

Table 1: Overview of the datasets used in this study. *# Annotators* refers to the number of annotations per comment.

Dataset	#Comments	#Annotators	#SDB dims	Task
Kumar [9]	106,035	5–6	10	Toxicity
Sap [8]	626	5–6	3	Hate Speech
DICES-350 [15]	72,103	60–70	4	Hate Speech
DICES-990 [15]	43,050	123	4	Hate Speech

We use four datasets from current Natural Language Processing (NLP) research that include SDB information at the level of individual annotators.[3] Specifically, we use the Kumar" [9] and Sap" [8] datasets, which are concerned with Toxicity and Hate Speech detection respectively, as well as the DICES-350 and DICES-990 datasets [15], which focus on LLM response safety. For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby framing the task as Hate Speech detection.

While the Toxicity and Hate Speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Consequently, the sources of annotator disagreement and bias that arise in these tasks are expected to be structurally similar, which makes them suitable for joint analysis within our framework.

The datasets differ substantially in scale and annotation design. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct SDB and personal-experience dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets, making them an invaluable resource for studying polarization and disagreement at scale.

[3]While we would like to test our framework on other research areas and domains, NLP remains the only large field to our knowledge that systematically investigates annotator biases and includes the necessary SDB information.

A brief overview of dataset characteristics is provided in Table 1. Across all datasets, we observe a non-trivial degree of polarization in most annotations, as illustrated in Figure 6. Finally, *Age* is represented as a continuous (ratio) variable in some datasets and as a discretized interval variable in others; we discuss the implications of this difference, as well as our own discretization strategy, in Appendix C.

4 Results

4.1 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided SDB dimensions can partially explain polarization in the toxicity/racism annotations. The results for the Kumar and Sap datasets can be found in Tables 7c and 2, while Tables 7a and 7b display the results for the DICES-350 and DICES-990 datasets respectively. We sample 20,000 comments from the Kumar dataset due to computational complexity.

4.2 Which factors contribute to polarization?

Table 2: Aposteriori unimodality results for the kumar dataset.

SDB	Value	apunim	support
Age	1) 18 - 24	-0.0090	9742
	2) 25 - 34	0.0324***	33203
	3) 35 - 44	-0.0093	21055
	4) 45 - 54	0.0006	11025
	5) 55 - 64	-0.0037	6123
	6) 65 or older	0.0001	2567
Education	1) Doctoral degree	0.0048	1019
	2) Professional degree	0.0038	1321
	3) Master's degree	0.0147**	13420
	4) Bachelor's degree	0.0401***	34085
	5) Associate degree	-0.0113**	9122
	6) College, no degree	-0.0335***	16666
	7) High School graduate	-0.0079*	7326
	8) No high school	0.0004	462
Ethnicity	Asian	-0.0019	5062
	Black	0.0110*	10999
	Hispanic	-0.0025	2389
	Multiracial	-0.0041	3682
	Other	0.0022	1507
	Unknown	-0.0030	1160
	White	-0.0377***	45306

Continued on next page

Table 2: Aposteriori unimodality results for the kumar dataset.

SDB	Value	apunim	support
Gender	Female	-0.0278***	40686
	Male	0.0275***	37929
	Nonbinary	-0.0008	489
	Unknown	-0.0029	621
Has Been Targeted	False	-0.0633***	45234
	True	0.0349***	24491
Is Parent	No	-0.0707***	37231
	Prefer not to say	-0.0023	831
	Yes	0.0535***	41493
Is Transgender	No	-0.0894***	11978
	Prefer not to say	0.0005	856
	Yes	0.0013	2306
Political Affiliation	Conservative	0.0331***	23010
	Independent	-0.0157***	22818
	Liberal	-0.0287***	32917
	Other	-0.0049	1822
	Prefer not to say	0.0015	3548
Religion Important	1) No	-0.0804***	26008
	2) Not very	-0.0154***	10055
	3) Somewhat	0.0210***	20169
	4) Very	0.0487***	27284
	Prefer not to say	0.0001	1179
Seen Toxicity	False	0.0311***	20185
	True	-0.0717***	43265
Sexual Orientation	Bisexual	0.0175***	10005
	Heterosexual	-0.0571***	38048
	Homosexual	-0.0017	2975
	Other	-0.0003	728
	Prefer not to say	-0.0017	1751
Technology Impact	1) Very negative	-0.0009	784
	2) Somewhat negative	-0.0080	7973
	3) Neutral	0.0027	10720
	4) Somewhat positive	-0.0265***	40477
	5) Very positive	0.0208***	22471
Toxicity Problem	1) Never	0.0058	4962
	2) Rarely	0.0177***	14457
	3) Occasionally	-0.0058	29476
	4) Frequently	-0.0217***	24774
	5) Very Frequently	-0.0145***	10656

(a) Aposteriori unimodality results for the dices-350 dataset.

SDB	Value	apunim	support
Age	1) Gen. X+	0.0186	9703
	2) Millenial	-0.0059	11268
	3) Gen. Z	-0.0066	17528
Educat.	College +	0.0124	23475
	High school -	-0.0152	12833
	Other	-0.0214*	2191
Gender	Man	-0.0193	19093
	Woman	0.0215	19406
Race	African Am.	-0.0428***	9077
	Asian	0.0659***	8138
	Latino	-0.0030	6886
	Multiracial	-0.0001	5008
	White	-0.0198	9390

(b) Aposteriori unimodality results for the dices-990 dataset.

SDB	Value	apunim	support
Age	1) Gen. X+	-0.0471***	14384
	2) Millenial	-0.0043	38436
	3) Gen. Z	0.0644***	13741
Educat.	College +	0.0095	59142
	High school -	-0.0751***	7419
Gender	Man	0.0278***	32494
	Woman	-0.0235***	33471
Race	African Am.	0.0909***	5810
	Asian	0.0100	18428
	Latino	-0.1328***	6610
	Other	0.0753***	24321
	White	-0.1196***	11392

(c) Aposteriori unimodality results for the sap dataset.

SDB Feature	Value	apunim	support
Age	1) Gen. X+	0.0192	743
	2) Millennial	0.0189	1792
	3) Gen. Z	-0.0153	239
Ethnicity	black	-0.2967***	1041
	white	0.1400***	2048
Gender	man	0.0942***	1635
	woman	-0.1810***	1333

Fig. 7: Aposteriori unimodality results across the dices-350, dices-990, and sap datasets.

Polarization is consistently attributed to annotator Ethnicity across all datasets, and especially so on hate-speech detection tasks

Across all four datasets, we observe a consistent pattern in which annotation polarization can be attributed to differences in annotator ethnicity, and in many datasets ethnic/racial effects are the strongest contributors to polarization. In each case, statistically significant effects emerge for at least one of the major ethnic groups represented in the data—most notably *White* and *African American* annotators—either individually or in combination. Additionally, while polarization attributable to ethnicity is present in all cases, it is substantially more pronounced in the Hate Speech datasets than in the Toxicity dataset. This suggests that *ethnic background constitutes a systematic source of annotator disagreement* rather than isolated noise, and is *more prevalent on tasks explicitly investigating racism* against those backgrounds.

Some dimensions exert an overall influence to the dataset

Polarization attribution across different SDB dimensions can take both positive and negative values, reflecting that some groups actively contribute to overall polarization, while others tend to mitigate or "absorb" it (see §2.3). For example, in the DICES-350 dataset (Table 7a), Asian annotators show a strong positive contribution to polarization ($apunim = 0.0659$), whereas African American annotators contribute negatively ($apunim = -0.0428$), effectively reducing overall polarization. Similarly, in the Kumar dataset (Table 2), female annotators decrease polarization ($apunim = -0.0278$) while male annotators increase it ($apunim = 0.0275$), and respondents who report having been previously targeted also increase polarization ($apunim = 0.0349$), in contrast to those who have not ($apunim = -0.0633$).

Importantly, these effects do not necessarily sum to zero, which suggests that *certain dimensions may exert a systematic influence* on the total polarization of the dataset. In other words, some annotator characteristics can consistently increase or decrease observed polarization.

Educated annotators tend to increase polarization

Education tends to have a moderate but consistent directional effect across all three datasets where it is present. The direction of this effect is consistent; the more educated an annotator is, the more polarized their ratings become. This may indicate that educated annotators are more likely to exhibit subjective and nuanced views.

The two predominant genders display symmetrical, opposing effects

Across all datasets, the two predominant genders exhibit consistent, opposing and symmetrical effects in polarization attribution, with one group typically amplifying and the other mitigating overall polarization. In DICES-350, female annotators slightly increase polarization ($apunim = 0.0215$) while male annotators reduce it (-0.0193), whereas in DICES-990 the direction flips, with men increasing (0.0278) and women decreasing (-0.0235) polarization. Similar patterns are observed in the Sap dataset, where male annotators elevate polarization (0.0942) and female annotators strongly absorb it (-0.1810), and in the Kumar dataset, with male annotators increasing (0.0275) and female annotators reducing (-0.0278) polarization. Nonbinary or unknown annotators consistently exhibit near-zero effects, which is likely caused by the relative low sample populations. These results suggest that while gender consistently shapes annotation polarization, the specific direction of its influence is dataset-dependent.

4.3 Investigating ordinal attributes

Analysis on annotated data frequently ignores the different kinds of measurements in studies using sociodemographic and LIKERT-style attributes. Researchers often treat all dimensions as *nominal/categorical* even though some are inherently ordinal (e.g., LIKERT-scale, education), and some continuous/ratio variables (e.g., age). Others attempt to find an "optimal point" which can be used to binarize continuous variables [**?**], but this approach is misguided. By using any artificial cut-off point, the resulting

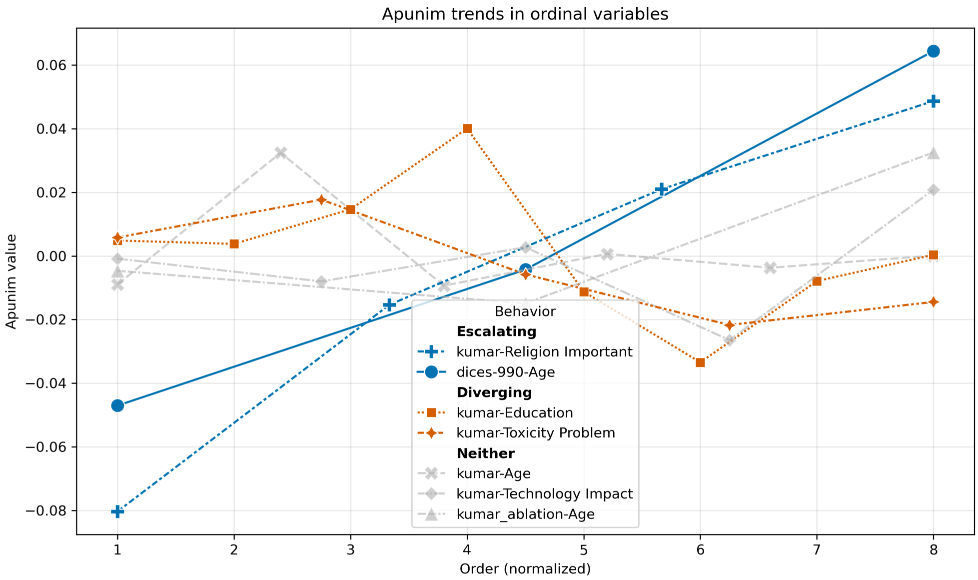

Fig. 8: The behavior of ordinal variables in all datasets. The x-axis represents the order of each factor (marked as `1), 2), ...` in Tables 7a,7b,2,7c). The x-axis is normalized to the smallest common denominator between the number of ordinal values.

analysis is immediately invalidated by any change in the underlying data, and it also discards valuable information not present in binary or nominal categories By treating these variables as categorical, not only do we lose valuable information, but we may magnify statistical errors [16].

Figure 8 shows that *retaining the order of ordinal attributes reveals interesting trends* that would be otherwise missed with categorical inference. We aggregate these trends in two patterns; *"Escalating"* and *"Diverging"*.

Diverging

Some ordinal attributes tend to cluster around a few dominant values; for example, *Age* and *Education* in the Kumar dataset. This pattern is expected because the Apunim algorithm is designed to identify clusters of annotations, causing nearby or similar values to be absorbed into the main effects driven by a few significant modes. Thus, it is common to find two halves of an ordinal scale being dominated by positive and negative apunim values respectively.

Escalating

However, some other attributes (*Age* in both DICES-350 and DICES-990, *Religion Importance* in the Kumar dataset) follow a strict, monotonic trend. This implies that, in some cases, the **amount of polarization increases the further apart an opinion is from neutrality**. For example, non-religious annotators are less likely to have polarizing opinions about what constitutes a toxic comment than the average annotator, while very religious annotators generate very polarizing judgments.

5 How many annotators?

While our test works best with many annotations per datapoint, the cost required for multiple human annotators is often prohibitive [11]. It is thus very important both DICES datasets, as well as the Sap dataset annotated by 100 distinct LLM annotators. While a more appropriate comparison would entail synthetic annotations in the DICES datasets themselves, we elect to use a standard task for LLM-as-a-judge (hate speech detection) instead of the relatively challenging and under-explored area of general LLM safety.

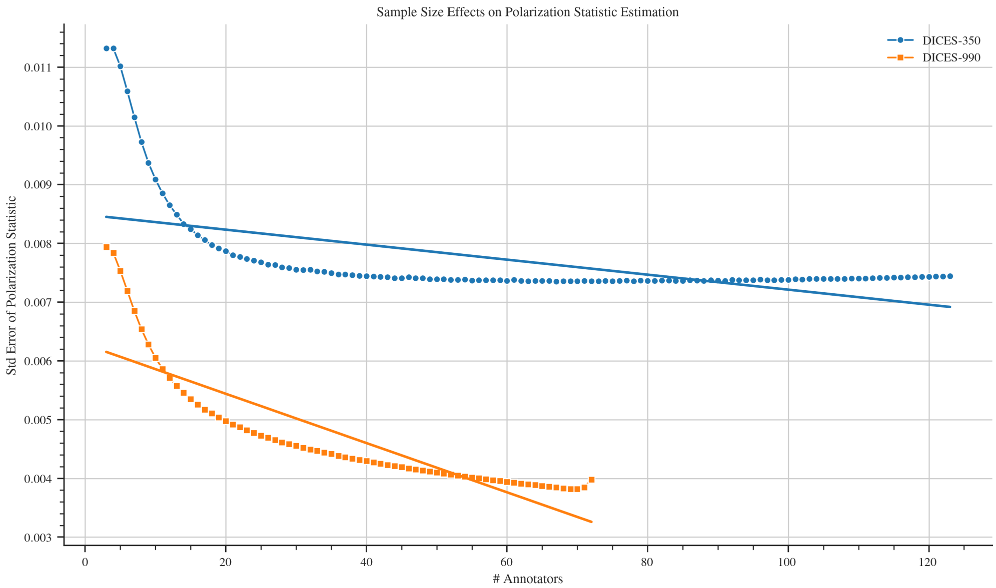

Fig. 9: Standard error of the *pol-statistic* when sampled with replacement for various number of annotators.

For each comment in all datasets, we randomly sample a subset of annotators and compute the apunim value. We repeat the procedure 30 times, and average all observations. This was a necessary step to reduce the variance inherent in random sampling. We also truncate the results for the DICES-350 dataset, since very few comments were annotated by the respective annotator counts.

Figure 9 demonstrates the effect of the number of annotators on the reliability of the polarization estimation. We can see that the metric becomes consistent with about 20 annotators per comment, after which we start to get diminishing results. This experiment was conducted on a subset of the Sap dataset, with LLM annotators having different SDBs.

6 Conclusion

We introduced the "Aposteriori Unimodality Test", a statistical test that detects whether certain sociodemographic groups partly cause polarization in annotations. Our test is resistant to low samples of annotations, bypasses common statistical issues such as sample dependence, and avoids newly-discovered issues such as the presence of inherent polarization and polarization direction. We apply our test two human-generated and two LLM-generated datasets and find interesting patterns on the former, and verify current literature on LLM SDB prompting on the latter.

7 Limitations

Since there are no other established tests that attribute polarization to sociodemographic characteristics, there is no way to verify that the results presented in §4 are accurate. Similarly, there is no qualitative way of evaluating our test, other than observations on the (non-) efficacy of LLM SDB prompting and our own intuition. Lastly, while our test seems to work sufficiently well with a small number of annotators per data-point, we still encourage researchers using this test to have at least a few annotators of each sociodemographic group that is being tested, in their dataset.

8 Ethical Considerations

While our work aims to help protect marginalized and disadvantaged groups by attributing polarization to certain subgroups, it can be taken advantage of by malicious actors. Given a dataset with fine-grained SDB information, these actors can use our test to target specific vulnerable groups. We thus urge researchers to keep datasets including such information protected, and provided to others only under explicit permission.

Appendix A Acronyms

LLM Large Language Model
SDB SocioDemographic Background
nDFU normalized Distance From Unimodality
FWER Family-Wise Error Rate
NLP Natural Language Processing

Appendix B Theoretical limitations

In §2.2 we compared the polarization of the whole set of annotations with the partitions by annotator group. This approach uses the same samples for both subset and superset, violating the fundamental hypothesis of independent samples. This also means the statistics for both will *always be similar to an extent.* This is especially pronounced in our case, due to the extreme overlap exhibited by tiny annotation sets. Additionally, when calculating p-values, the non-independence of samples induces an intrinsic, non-zero covariance between any statistic s calculated on A and on S.

A standard method of side-stepping this issue is to compare the subset with its complement to the superset. This can not be applied to our case. Consider the example in Figure 3; if we compared $nDFU_{men}$ with $nDFU_{women}$, the result would be very close to zero, indicating no polarization attribution to gender.

Appendix C How do we handle continuous variables?

Continuous attributes (such as age) can be discretized during analysis; this practice should generally be avoided [16, 17], although it is necessary when the data has extremely skewed distributions [16], the underlying data are already in discretized form, or when the tools used for analysis require some form of categorization. Our methodology is built upon nDFU, and thus requires histograms to be well-defined. Additionally, the data for Age is already discretized for the DICES and Kumar datasets; for the sake of comparison, we discretize the Sap dataset with the same method as the DICES dataset, instead of using generalized binning techniques based on statistics [18–20].

In order to make sure our results are not biased, we conduct ablation experiments with all ordinal variables from the Kumar dataset discretized to three factors. We observe that...

References

[1] United Nations. Targets of hate (2025). URL https://www.un.org/en/hate-speech/impact-and-prevention/targets-of-hate. Accessed: 2025-09-22.

[2] Dioum, S. What explains the increase in online hate speech? (2025). URL https://www.ucdavis.edu/magazine/what-explains-increase-online-hate-speech. Accessed: 2025-09-22.

Table C1: Aposteriori unimodality results for the kumar_ablation dataset.

SDB Feature	Value	apunim	support
Age	1) Gen. X+	-0.0047	8690
	2) Millennial	-0.0149**	31470
	3) Gen. Z	0.0325***	40951
Education	Neutral	0.0150***	13919
	No	0.0000	2340
	Yes	-0.0376***	39926
Religion Important	Neutral	0.0226***	21348
	No	-0.1111***	35028
	Yes	0.0485***	27284
Technology Impact	Neutral	0.0026	10720
	No	-0.0086*	8757
	Yes	0.0114*	42443
Toxicity Problem	Neutral	-0.0060	29476
	No	0.0255***	19334
	Yes	-0.0399***	34645

[3] Quaranto, A. Dog whistles, covertly coded speech, and the practices that enable them. *Synthese* **200**, 330 (2022).

[4] Sap, M., Card, D., Gabriel, S., Choi, Y. & Smith, N. A. Korhonen, A., Traum, D. & Màrquez, L. (eds) *The risk of racial bias in hate speech detection*. (eds Korhonen, A., Traum, D. & Màrquez, L.) *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 1668–1678 (Association for Computational Linguistics, Florence, Italy, 2019). URL https://aclanthology.org/P19-1163/.

[5] Kralj Novak, P. *et al.* Ciucci, D. *et al.* (eds) *Handling disagreement in hate speech modelling*. (eds Ciucci, D. *et al.*) *Information Processing and Management of Uncertainty in Knowledge-Based Systems*, 681–695 (Springer International Publishing, Cham, 2022).

[6] Pavlopoulos, J. & Likas, A. Distance from unimodality for the assessment of opinion polarization. *Cognitive Computation* **15**, 731–738 (2023). URL https://doi.org/10.1007/s12559-022-10088-2.

[7] Pavlopoulos, J. & Likas, A. Graham, Y. & Purver, M. (eds) *Polarized opinion detection improves the detection of toxic language*. (eds Graham, Y. & Purver, M.) *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 1946–1958 (Association for Computational Linguistics, St. Julian's, Malta, 2024). URL https://aclanthology.org/2024.eacl-long.117/.

[8] Sap, M. *et al.* Carpuat, M., de Marneffe, M.-C. & Meza Ruiz, I. V. (eds) *Annotators with attitudes: How annotator beliefs and identities bias toxic language detection.* (eds Carpuat, M., de Marneffe, M.-C. & Meza Ruiz, I. V.) *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 5884–5906 (Association for Computational Linguistics, Seattle, United States, 2022). URL https://aclanthology.org/2022.naacl-main.431/.

[9] Kumar, D. *et al.* *Designing toxic content classification for a diversity of perspectives*, SOUPS'21 (USENIX Association, USA, 2021).

[10] Tsirmpas, D., Androutsopoulos, I. & Pavlopoulos, J. Scalable evaluation of online facilitation strategies via synthetic simulation of discussions (2025). URL https://arxiv.org/abs/2503.16505. arXiv:2503.16505.

[11] Rossi, L., Harrison, K. & Shklovski, I. The problems of llm-generated data in social science research. *Sociologica* **18**, 145–168 (2024). URL https://sociologica.unibo.it/article/view/19576.

[12] Student. The probable error of a mean. *Biometrika* 1–25 (1908).

[13] Holm, S. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* **6**, 65–70 (1979). URL http://www.jstor.org/stable/4615733.

[14] Chen, S., Feng, Z. & Yi, X. A general introduction to adjustment for multiple comparisons. *Journal of Thoracic Disease* **9**, 1725–1729 (2017). Doi: 10.21037/jtd.2017.05.34; PMID: 28740688; PMCID: PMC5506159.

[15] Aroyo, L. *et al.* Oh, A. *et al.* (eds) *Dices dataset: Diversity in conversational ai evaluation for safety.* (eds Oh, A. *et al.*) *Advances in Neural Information Processing Systems*, Vol. 36, 53330–53342 (Curran Associates, Inc., 2023). URL https://proceedings.neurips.cc/paper_files/paper/2023/file/a74b697bce4cac6c91896372abaa8863-Paper-Datasets_and_Benchmarks.pdf.

[16] Streiner, D. L. Breaking up is hard to do: the heartbreak of dichotomizing continuous data. *Can J Psychiatry* **47**, 262–266 (2002).

[17] Andrade, C. Age as a variable: Continuous or categorical? *Indian J Psychiatry* **59**, 524–525 (2017).

[18] Sturges, H. A. The choice of a class interval. *Journal of the American Statistical Association* **21**, 65–66 (1926). URL https://doi.org/10.1080/01621459.1926.10502161.

[19] Scott, D. W. On optimal and data-based histograms. *Biometrika* **66**, 605–610 (1979). URL https://doi.org/10.1093/biomet/66.3.605.

[20] Freedman, D. & Diaconis, P. On the histogram as a density estimator:l2 theory. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **57**, 453–476 (1981). URL https://doi.org/10.1007/BF01025868.